

Acceptance Criteria — HL Sales & Receivables Management App

Scope: single-user internal app for managing Customers, Products, Transactions (Bon), receivables (Piutang), bonuses, and reporting for the business "HL". Currency: **IDR (Rp)** only, **no tax/PPN**. "jt" = juta = 1,000,000. Accounting basis: **cash basis** — omzet, profit, and bonus eligibility are recognized only when a transaction is **Lunas** (paid). See Decision D3.

Glossary / Conventions

Term	Meaning
Bon	A single transaction / invoice (identified by Nomor Bon).
LM / BR	Product types. Each customer has a separate discount set per type.
Harga Modal	Product cost price (what HL pays). Used for profit only.
Harga Base / Harga Jual	Product list price before discount.
Diskon bertingkat (cascading)	A sequence of % discounts applied one after another, NOT summed.
Ongkir	Shipping cost. Pass-through — charged to customer, no profit impact.
Omzet	Revenue = discounted price × qty (shipping excluded). Recognized when Lunas.
Laba HL	Profit = (discounted price – modal) × qty. Recognized when Lunas.
Piutang	Receivable / unpaid amount. Default status of a new transaction.
Lunas	Paid / settled.

1. Authentication

- **AC-1.1** The app requires login before any feature is accessible.
 - **AC-1.2** Exactly one user account exists; there is no self-registration flow.
 - **AC-1.3** Given valid credentials, the user is logged in and lands on the home/dashboard.
 - **AC-1.4** Given invalid credentials, login is rejected with a clear error and no access is granted.
 - **AC-1.5** A logged-in session persists until logout (or session expiry, if implemented), and a logout option is available.
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2. Customer Management (CRUD)

A Customer has: **Nama**, **Diskon per tipe (LM & BR)**, and a **Bonus eligibility threshold**.

- **AC-2.1** User can **create** a customer with a name (required).
- **AC-2.2** User can **edit** any field of an existing customer.
- **AC-2.3** Deleting a customer performs a **soft-delete**: the customer is hidden from new selections, but all historical transactions remain intact and visible in reports.
- **AC-2.4** Each customer holds **two independent discount sets**: one for **LM**, one for **BR**.
- **AC-2.5** A discount set is an **ordered list** of percentage values (e.g. LM = [20, 20, 10]). Order matters because discounts are applied sequentially.
- **AC-2.6** Within a discount set, the user can **add**, **edit**, and **delete** individual discount steps.
- **AC-2.7** Discount values must be numeric and between 0 and 100; invalid entries are rejected.
- **AC-2.8** Each customer has a **bonus eligibility threshold** (a Rupiah amount, e.g. 10,000,000) used by the bonus logic in §5.

Cascading discount rule (enforced everywhere a price is shown/calculated):

Given base price B and discount steps $[d1, d2, \dots, dn]$ (in %), discounted unit price = $B \times (1 - d1/100) \times (1 - d2/100) \times \dots \times (1 - dn/100)$.

- **AC-2.9** Example check: $B = 100$, LM [20, 20, 10] $\rightarrow 100 \times 0.8 \times 0.8 \times 0.9 = 57.6$. The effective discount is **42.4%**, and the system must **not** treat it as 50%.

3. Product Management (CRUD)

A Product has: **Nama**, **Harga Modal** (cost), **Harga Base/Jual** (selling price), **Tipe (LM or BR)**.

- **AC-3.1** User can create, edit, and delete products.
 - **AC-3.2** Tipe is restricted to **LM** or **BR**.
 - **AC-3.3** Harga Modal and Harga Base are numeric and ≥ 0 .
 - **AC-3.4** Harga Modal is used only for profit (Laba) calculations and is never shown as a customer-facing price.
 - **AC-3.5** Deleting a product performs a **soft-delete**: hidden from new selections, history preserved.
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4. Transaction (Bon) Management

Each transaction captures:

- **Tanggal** — defaults to today, editable.
- **Nomor Bon** — receipt number, **must be unique**.
- **Customer** — chosen from existing customers.
- **Produk line items** — each: product (from catalog), quantity, with BR/LM type and price shown.
- **Ongkir** — shipping amount.
- **Deskripsi** — free text.
- **Bonus** — on/off toggle (see §5).
- **Status** — Piutang / Lunas; defaults to **Piutang**.
- **AC-4.1** Date field is pre-filled with the current date and can be changed.
- **AC-4.2** Nomor Bon is required and **must be unique**; saving a duplicate Nomor Bon is rejected with a clear error.
- **AC-4.3** Customer is selected from the existing customer list (not free text).
- **AC-4.4** Products are selected from the existing product catalog (not free text).

- **AC-4.5** A transaction supports **multiple product lines**, each with its own quantity (≥ 1).
- **AC-4.6** For each product line, the UI shows the product **type (LM/BR)** and the **price applied** for this customer (the discounted unit price computed from that customer's discount set for that product's type).
- **AC-4.7** The discount applied to each line is derived automatically from **the selected customer × the product's type** — the user does not type discounts manually on the transaction.
- **AC-4.8** Ongkir is numeric and ≥ 0 and is captured per transaction (not per line).
- **AC-4.9** Status defaults to **Piutang** on creation; user may set Lunas later (see §6 flows).
- **AC-4.10** User can view, edit, and delete a transaction.
 - **AC-4.10.1** Editing a transaction recalculates omzet, profit, and totals.
- **AC-4.11** The transaction shows computed values: **per-line omzet**, **transaction omzet (excl. ongkir)**, **ongkir**, and **total amount owed = omzet + ongkir**.

Calculation rules (per transaction):

- Line discounted unit price = cascading discount (§2) using the customer's set for that line's type.
- Line omzet = discounted unit price × quantity.
- Transaction omzet = Σ line omzet (ongkir **excluded**).
- Amount owed (Piutang) = transaction omzet + ongkir.
- Line Laba HL = (discounted unit price – harga modal) × quantity.
- Ongkir is **pass-through** → it does **not** affect Laba HL.
- Omzet and Laba are **recognized only when the transaction becomes Lunas** (cash basis).

5. Bonus Logic

A "bonus bon" is a transaction opened specifically to give a customer free bonus products once they have earned it through accumulated **paid** omzet.

- **AC-5.1** Each customer has a **bonus eligibility threshold** (§2.8), e.g. Rp 10,000,000.
- **AC-5.2** The system maintains a **running accumulated omzet** per customer, counting

only Lunas (paid) transactions.

- **AC-5.3 Bonuses stack:** the number of bonuses earned = $\text{floor}(\text{accumulated paid omzet} / \text{threshold})$ minus bonuses already granted.
- **AC-5.4** When a customer has at least one earned bonus, the system surfaces eligibility (flag/notification) and the number of bonuses available.
- **AC-5.5** A bonus is recorded as a transaction with **Bonus = on**. The user may include **multiple bonuses in a single bon**.
- **AC-5.6** Each bonus granted **consumes one threshold's worth** of accumulated omzet; the **remainder carries over** to the next bonus cycle.
- **AC-5.7** Bonus product lines are given **free**: they are **excluded from omzet** and their cost **does not reduce Laba HL** (bonus cost is ignored in profit).
- **AC-5.8** Bonus transactions are clearly distinguishable from normal sales in lists and recaps so they don't inflate revenue/receivables.

Scenario (worked example):

```
Given Customer A threshold = 10,000,000
And Customer A accumulated PAID omzet = 25,000,000
And no bonuses granted yet
Then 2 bonuses are available (floor(25/10) = 2)
When the user creates one bonus bon and assigns both bonuses
Then 20,000,000 is consumed (2 × threshold)
And 5,000,000 carries over toward the next bonus
And the bonus products are free → no omzet, no profit impact
```

6. Customer Detail Page

Each customer has a dedicated page showing their activity, grouped **by month**.

- **AC-6.1** The page lists the customer's transactions grouped by month (selectable by month/year).
- **AC-6.2** Selecting a month shows, for that month:
 - List of transactions (Bon) with date, Nomor Bon, status, amount.
 - **Total Piutang** (sum of amounts owed on Piutang transactions = omzet + ongkir).
 - **Total sudah dibayar** (sum of amounts paid on Lunas transactions = omzet +

ongkir).

- **Total Omzet** (Σ omzet of Lunas transactions, excl. ongkir).
- **Total Laba HL** (Σ profit of Lunas transactions).
- **AC-6.3** Omzet is shown with **BR and LM in separate columns** (plus a combined total).
- **AC-6.4** User can **view** and **download (PDF)** these lists (Piutang list, transaction list).

Settlement (Pelunasan) flows:

- **AC-6.5 — Settle a whole month:**

```
Given the user is viewing a month (e.g. January) on a customer's page
When the user clicks "Sudah Lunas" (settle month)
Then a modal asks for a Tanggal Pelunasan (payment date)
When the user confirms
Then every transaction in that month for that customer is set to status Lunas
    And each carries the entered payment date
    And their omzet/profit become recognized (cash basis)
```

- **AC-6.6 — Settle a single Bon:**

```
Given the user opens a single transaction (Bon) detail
When the user clicks "Lunas"
Then the same payment-date modal appears
When the user confirms
Then only that transaction is set to Lunas with the entered payment date
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- **AC-6.7** Settling updates totals immediately (Total Piutang ↓, Total sudah dibayar ↑, recognized Omzet/Laba ↑, accumulated bonus omzet ↑).
- **AC-6.8** Already-Lunas transactions are not re-settled and are visually distinct.
- **AC-6.9** Clicking a Bon opens its full detail (lines, qty, prices, ongkir, omzet, status, payment date if any).

7. Recap / Reporting

- **AC-7.1** Recap available **per customer**.
- **AC-7.2** Recap available **per product type (LM / BR)**.
- **AC-7.3** Recap available **overall** (all customers combined).

- **AC-7.4** Each recap can be filtered/grouped **per month** and **per year**.
- **AC-7.5** Each recap reports at minimum: **Total Omzet** (Lunas), **Total Laba HL** (Lunas), **Total Piutang (outstanding)**, **Total sudah dibayar**, broken down by **LM vs BR** where relevant.
- **AC-7.6** Overall recap shows **total Laba HL across all customers**.
- **AC-7.7** Bonus transactions are excluded from omzet/revenue/profit totals (per §5.7–5.8) and may be reported separately as a bonus log.
- **AC-7.8** Recaps are **downloadable as PDF**.

8. Master Calculation Reference (single source of truth)

Quantity	Formula
Discounted unit price	$\text{Base} \times \prod (1 - d_i/100)$ over the customer's discount steps for that type
Line omzet	$\text{discounted unit price} \times \text{qty}$
Transaction omzet	$\sum \text{line omzet}$ (ongkir excluded)
Amount owed (Piutang)	$\text{transaction omzet} + \text{ongkir}$
Line Laba HL	$(\text{discounted unit price} - \text{harga modal}) \times \text{qty}$
Transaction Laba HL	$\sum \text{line Laba HL}$ (ongkir excluded — pass-through)
Recognized Omzet (reports)	$\sum \text{transaction omzet where status} = \text{Lunas}$
Recognized Laba HL (reports)	$\sum \text{transaction Laba HL where status} = \text{Lunas}$
Total paid	$\sum (\text{omzet} + \text{ongkir}) \text{ where status} = \text{Lunas}$
Total outstanding piutang	$\sum (\text{omzet} + \text{ongkir}) \text{ where status} = \text{Piutang}$
Bonus accumulator	$\sum \text{omzet where status} = \text{Lunas}$ (per customer)
Bonuses available	$\text{floor}(\text{bonus accumulator} / \text{threshold}) - \text{bonuses already granted}$

Bonus items	free → 0 omzet, 0 profit impact
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9. Confirmed Decisions (resolved)

#	Question	Decision
D1	Ongkir & profit	Pass-through — shipping adds no profit. Laba = omzet – modal.
D2	Receivable vs omzet	Customer owes omzet + ongkir ; omzet excludes ongkir .
D3	Omzet / eligibility basis	Only Lunas (paid) transactions count → cash basis for omzet, profit, and bonus accumulation.
D4	Bonus mechanics	Bonuses stack ; multiple bonuses can be placed in one bon ; each consumes one threshold, remainder carries.
D5	Bonus product cost	Ignored in profit — free bonus items do not reduce Laba HL.
D6	Deleting items with history	Soft-delete (hide from new use, keep history).
D7	Nomor Bon	Must be unique ; duplicates rejected.
D8	Export format	PDF .
D9	Currency / tax	IDR only, no tax/PPN .